Poster 597; Topic 04 Affektive Störungen

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Researching Depression

Elevated and Stable Arousal and Wakefulness Regulation in Major Depressive Disorder: Positive Association with Depression Severity

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INTRODUCTION

Background:

- An upregulated brain arousal has been suggested as a possible pathogenetic factor for major depressive disorder (MDD), as shown by resting-state EEG studies and clinical symptomatology (e.g., high inner tension and long sleep-onset latencies).
- The Arousal and Wakefulness Regulation Scale (AWRS) provides a less-resource intensive alternative to measure the self-reported arousal and wakefulness regulation (AWR), and it has not been evaluated in populations with MDD.

Research questions:

1) Is depression severity associated with an upregulation of arousal and wakefulness regulation (AWR), as indexed by higher AWRS score?
2) Are changes in AWR, as indexed by changes in AWRS scores, associated with changes in depression severity?

METHODS

Recruitment in five German university clinics: *Medizinische Hochschule Hannover, Universitätsklinikum Schleswig-Holstein, Universitätsmedizin Greifswald, Universitätsklinikum Würzburg*, and *Universitätsklinikum Frankfurt am Main* (Data collection period: January 2024 – October 2026)

Participants characteristics:

- age between 18 and 70 years old
- inpatients with moderate-to-severe MDD (MADRS ≥ 20)
- first depressive episode before 50 years old

Baseline (Day 1 – 7)
N = 638
Instruments: AWRS & HDRS-24

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Follow-up (Day 49 ± 3)
N = 380
Instruments: AWRS, HDRS-24

Statistical Analyses:

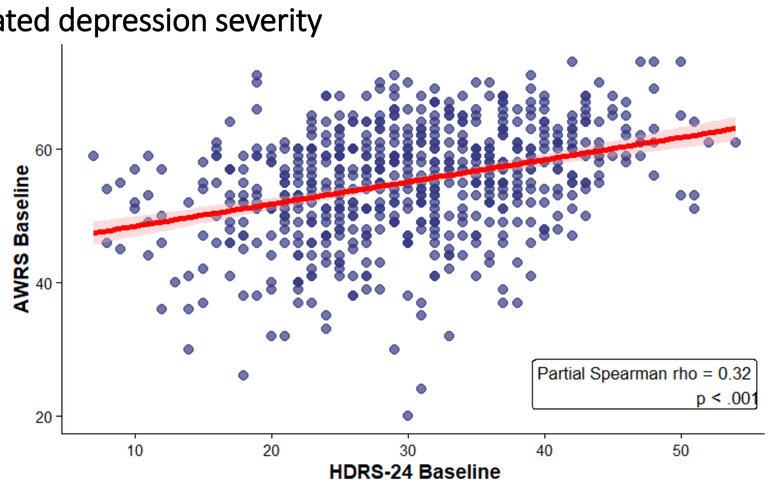
- **Partial spearman correlation** controlled for age and gender
- Change scores were the difference between follow-up and baseline

RESULTS

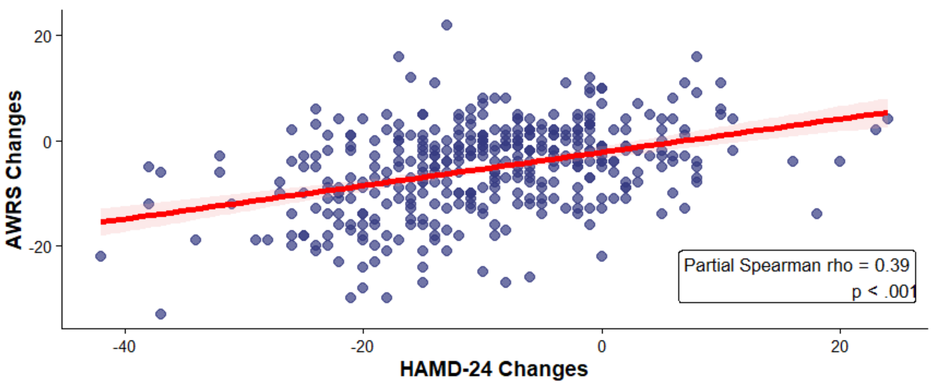
Sample Characteristics:

- Mean age: 38.2 (SD = 13.3)
- Gender: 298 men, 330 women, 10 non-binary

Result 1: Elevated and stable AWR (higher AWRS-score) correlates with elevated depression severity



Result 2: Reduction in AWRS correlates with reduction in depression severity



DISCUSSION AND IMPLICATIONS

- Elevated and stable AWR could reflect a state-dependent correlate of depression
- Comparison of euthymic people with a lifetime diagnosis of MDD to healthy controls is necessary to determine whether upregulated AWR is a state-modulated trait
- Future studies should explore if self-reported AWR is more strongly associated with specific depression symptoms (e.g., insomnia) than with others (e.g., changes in appetite)

References:
Heer, A., Reich, H., Sander, C., & Hegerl, U. (2025). *Too tired or too wired? Construction and Initial Validation of the Arousal and Wakefulness Regulation Scale (AWRS)* (Vgz9m_v1). PsyArXiv. https://doi.org/10.31234/osf.io/vgz9m_v1
Hegerl, U., & Hensch, T. (2014). The vigilance regulation model of affective disorders and ADHD. *Neuroscience & Biobehavioral Reviews, 44*, 45-57.
Ulke, C., Tenke, C. E., Kayser, J., Sander, C., Böttger, D., Wong, L. Y. X., Alvarenga, J. E., Fava, M., McGrath, P. J., Deldin, P. J., Mcinnis, M. G., Trivedi, M. H., Weissman, M. M., Pizzagalli, D. A., Hegerl, U., & Bruder, G. E. (2019). Resting EEG Measures of Brain Arousal in a Multisite Study of Major Depression. *Clinical EEG and Neuroscience*, *50*(1), 3–12. https://doi.org/10.1177/1550059418795578
Xie, M., Huang, Y., Cai, W., Zhang, B., Huang, H., Li, Q., Qin, P., & Han, J. (2024). Neurobiological Underpinnings of Hyperarousal in Depression: A Comprehensive Review. *Brain Sciences*, *14*(1), 50. https://doi.org/10.3390/brainsci14010050

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